VISIBLE LIGHT INDOOR POSITIONING FASHIONED WITH A SINGLE TILTED OPTICAL RECEIVER

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ABSTRACT

We propose a compact visible light indoor positioning system fashioned with a single transmitter and a single tilted receiver. The tilted optical receiver is mounted on a rotatable platform and thus, various azimuth angles can be obtained by rotating the platform, which offers a feasible way to perform multiple measurements with different azimuth angles to achieve the angle gain. An indoor positioning algorithm is provided in comparison with the angle gain difference. Experimental results show that the average distance error between the estimated position and the real position is less than 25 mm in an indoor environment of 600mm×600mm×1135mm, which is in good agreement with the simulations. This work indicates that the proposed compact system is promising for visible light indoor positioning.

Keywords: *Light-emitting diodes, Indoor positioning, Optical communications, Algorithm*

1. INTRODUCTION

Visible light communications (VLC) using a light emitting diode (LED) is not only for illumination but also for data transmission [1-4]. As a promising technique for wireless communications, the advantages of VLC are muti-fold, including energy conservation, the absence of mercury, additional brightness, extended lifetime, high security, the absence of electromagnetic interference, and a higher data rate comparing with other wireless communications. VLC has been verified as being suitable for various applications such as indoor localization [5-6], heterogenous networks [7], intelligent transportation [8-9] and broadcasting [10-11].

The LED has a certain radiation pattern, and the measured radiation intensity depends on the longitudinal and azimuth angle and the distance from the LED. Hence, the distance from the LED can be determined by the gain differences, which offers a promising way to realize indoor positioning by the utilization of visible light. Multiple LED transmitters or optical receivers are often used to achieve the gain difference for indoor localization. However, in the case of multiple transmitters used in the indoor positioning system, inter-cell interference may be a serious problem in a single channel system, which uses the same modulation frequency band [12-13]. The transmission time of signals need to be managed to avoid overlap of received signals without the inter-cell interference. The carrier allocation technique [14] or tilting the receiver plane to collect the maximum optical power [15] are demonstrated to overcome the inter-cell interference to improve the performance of the VLC system. Yang et. al proposed a single transmitter and multiple tilted optical receivers for visible light indoor localization [16-17]. The angle gain difference that is a function of the incidence angle and the coordinates of the transmitter and the receiver offers a 3-D positioning algorithm. The inter-cell interference is thus avoided. A minimum separation distance between the photo diodes (PDs) is required for obtaining the difference of the received signal power and thus, the size of the optical receiver becomes larger. Since the positioning algorithm is provided by the angle gain difference, a single tilted optical receiver is promising for the compact receiver, where angle gain difference can be realized by rotating the tilted optical receiver.

Here, we propose and experimentally demonstrate a visible light indoor positioning system fashioned with a single transmitter and a single tilted optical receiver. Both the inter-cell interference in the transmitters and the influence of the devices in the receivers are managed. The azimuth angle varies by rotating the tilted optical receiver and thus, angle gain difference is achieved, resulting in a positioning algorithm. Experimental results show that the average distance error between the estimated position and the real position is less than 25 mm in an indoor environment of 600mm×600mm×1135mm, which is in good agreement with the simulations.

2. PROPOSED SYSTEM MODEL

Figure 1 shows the schematic of the proposed indoor positioning system, where the LED transmitter locates on the center of the ceiling facing to the floor, and the

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receiver is placed on the floor. Let V_t and V_r be the vector of the transmitter and the receiver, respectively, while V_{rt} is the vector from the receiver to the transmitter. α is the angle of irradiance from the transmitter and β is the incidence angle of the receiver. The optical receiver is fixed on a tilted stage, which is connected with a rotation platform that works in the XY-plane. The polar angle θ is an angle between V_r and the normal vector of the XY-plane, which is equal to the inclination angle$\beta$of the receiver. And the azimuth angle φ is an angle between the X-axis and the projection of V_r in the XY-plane, which is determined by the position and the rotation angle ε of the tilted receiver. In this geometry, the polar angle θ and the azimuth angle φ are equal to the inclination angle δ and the rotation angle ε, respectively. Hence, it is feasible to obtain different polar angle θ and azimuth angle φ by tilting and rotating the receiver plane.

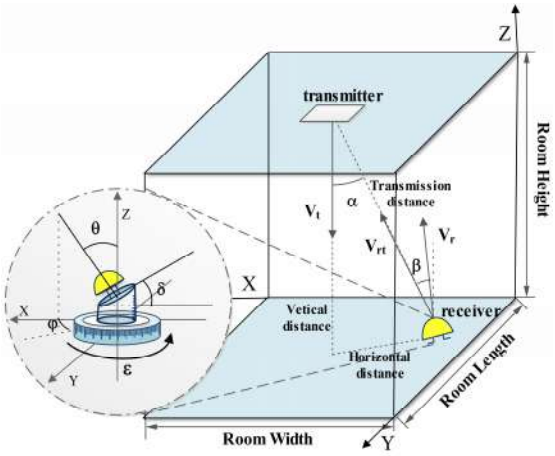


Fig. 1. The proposed indoor positioning system fashioned with a single transmitter and a single tilted receiver.

The radiation angle α and the incidence angle β are the same when the transmitter plane is parallel to the receiver plane. Different receiver planes correspond to different incidence angle β, which can be tuned by tilting the receiver plane. The incidence angle β is given as

$$\beta = \cos^{-1}\left(\frac{(V_r, V_{rt})}{D}\right) = \cos^{-1}\left(\frac{(a-x)\sin\theta\cos\varphi + (b-y)\sin\theta\sin\varphi + (c-z)\cos\theta}{\sqrt{(a-x)^2+(b-y)^2+(c-z)^2}}\right) \quad (1)$$

where (V_r, V_{rt}) is the inner product of V_r and V_{rt}, V_r becomes [$sin\theta cos\varphi$, $sin\theta sin\varphi$, $cos\theta$] after tilting the receiver plane and D is the transmission distance. (a, b, c) and (x, y, z) are the coordinates of the transmitter and the receiver, respectively, which are constant for a given transmitter and receiver. Therefore, the incidence angle β is the function of the polar angle θ and the azimuth angle φ. If the polar angle θ is set to be a constant by fixing the inclination angle δ of the optical receiver, the incidence angle β is thus determined by the azimuth angle φ. Rotating the platform generates different azimuth angle φ, resulting in different incidence angle β.

For simplicity, the received optical power in our case is assumed to be in a line-of-sight (LOS) environment. The received RF power of the optical wireless channel in a LOS environment can be expressed as

$$P_{RF} = (P_t^2 / D^4) C_r G_t^2(\alpha) G_r^2(\beta) \quad (2)$$

Where P_{RF} and P_t represent received RF power and average transmitted optical power, respectively. D is the transmission distance, and C_r is the RF power constant that is a combination of the optical-to-electrical conversion efficiency and the optical power constant. Since only a single optical receiver is used in the proposed indoor positioning system, C_r is constant. $G_t(\alpha)$ and $G_r(\beta)$ are the transmitter and receiver gain according to α and β, respectively. P_t and $G_t(\alpha)$ are constant for a given transmitter and receiver. Hence, the received RF power difference is determined by $G_r(\beta)$, which can be expressed as

$$G_r(\beta) = \exp(-\beta^{S_r} / K_r) \quad (3)$$

S_r is a slope constant as lens shape of receiver and K_r is related to semi-angle $\beta_{1/2}$ at the half power of optical receiver which can be expressed as $K_r = -(\beta_{1/2})S_r / ln2$. Since $\beta_{1/2}$ is constant, $G_r(\beta)$ is the function of β that is determined by the azimuth angle φ, thus, every gain distribution corresponds to an azimuth angle φ, which can be tuned by rotating the platform. Figures 2(a)-(c) demonstrate the simulated gain distribution according to the indoor position at the azimuth angle φ of 0°, 90°, 180°, respectively, where the polar angle is 10°, the indoor environment is 600mm×600mm×1135mm, the height of the optical receiver is 97.3mm, and the transmitter coordinates are (300, 300, 1135). It can be seen that the gain distributions are different due to the difference in the azimuth angle φ.

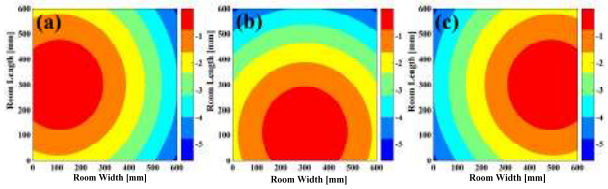


Fig. 2. The incidence angle gain distribution at the azimuth angle φ of 0° for (a), 90° for (b), 180° for (c), where the polar angle θ is 10°.

Since the received RF power P_{RF} corresponds to the angle gain distribution $G_r(\beta)$, the angle gain difference can be obtained by comparing with different received RF powers P_{RF}, which are measured by spectrum analyzer. The gain difference is expressed as

$$\Delta P_{ij} = P_{iRF} - P_{jRF} \approx \Delta G_{ij} = G_{ir}(\beta_i) - G_{jr}(\beta_j). \quad (4)$$

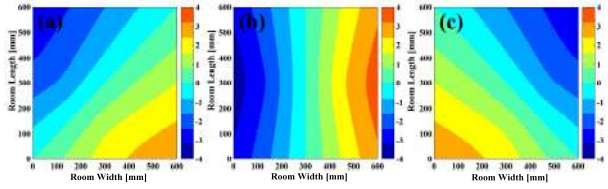


Fig. 3. The incidence angle gain difference of the optical receiver in the proposed system between different azimuth angles φ (a) 0° and 90°, (b) 0° and 180°, (c) 90° and 180°, respectively.

P_{iRF} is the ith received RF power and $G_{ir}(\beta_i)$ is the ith receiver gain. The incidence angle gain difference

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distribution according to the indoor position is thus generated from the incidence angle gain distribution. Figures 3(a)-(c) show the incidence angle gain difference of the optical receiver between different azimuth angles φ: (a) 0° and 90°, (b) 0° and 180°, (c) 90° and 180°, respectively, where the polar angle θ is 10°. Each gain difference value corresponds to one particular area of the environment and offers a positioning algorithm.

2. EXPERIMENT AND RESULTS

Figure 4(a) illustrates the experimental environment of the proposed visible light indoor positioning system. A single white LED is employed to illuminate the room with a launching power of 1W. The optical receiver is mounted on one 600mm×600mm optical breadboard. The distance between the LED transmitter and the optical breadboard is 1135mm and the height of the receiver is 97.3mm. Hence, the coordinates of the transmitter are defined as (300, 300, 1135) and the z-coordinate of the receiver is 97.3. Figure 4(b) shows the zoom-in image of the tilted optical receiver. A commercially available Si PIN PD (Hamamatsu S6968) is used to realize the optical-electrical conversion. The photocurrent that is generated by Si PIN PD is firstly amplified by a transimpedance amplifier, which converts the current into voltage in a wide band. The low frequency noise is managed by a high-pass filter. Then, a high-speed low-noise amplifier in association with a low-pass filter is used to obtain the amplified signal output with low noise interference. The process can efficiently reduce the negative influences from the ambient light and the power supply. The module that can supply a stable and repeatable testing environment for the experiment is fixed on a tilted stage with a tilted angle of 10°. The tilted stage is connected to a Thorlabs CRM1P rotation mount with a resolution of 5 arcmin.

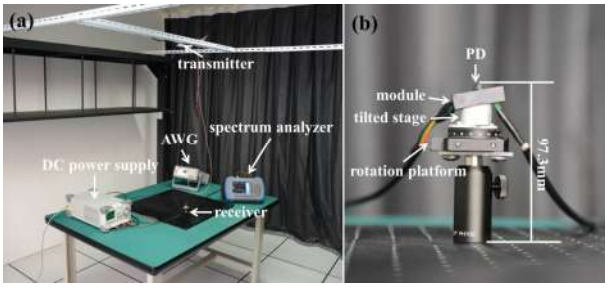


Fig. 4. (a) The experimental environment, (b) the tilted optical receiver.

Constant bias voltage and the sine signal with the frequency of 200kHz that are realized by an Agilent 33522A arbitrary waveform generator (AWG) are applied to the LED transmitter through a bias-Tee. Multiple measurements are carried out at different azimuth angles by rotating the tilted receiver plane, and the received signal powers are measured by an Agilent N9342C handhold spectrum analyzer. The received RF power is defined as the peak value of the spectrum analyzer and the power difference can be obtained in comparison with different measured RF powers. Figures 5(a)-(c) illustrate the measured signal power at the different azimuth angle φ of 0°, 90° and 180°, respectively. The real position of the optical receiver is (400, 400), and the measured signal powers P_1, P_2 and P_3 are -51.91dBm, -51.92dBm and -50.53dBm, respectively. Hence, the gain differences of ΔP_{21}, ΔP_{23} and ΔP_{31} are -0.01, -1.39, 1.38dB, respectively.

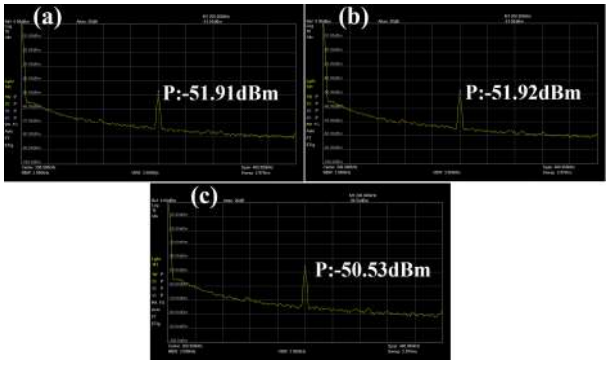


Fig. 5. The measured signal power at the different azimuth angle φ: (a) for 0°, (b) for 90°, and (c) for 180°.

Since the signal power difference is determined by the incidence angle gain difference, the possible position of the optical receiver can be obtained by a combination of the measured signal power difference and the calculated angle gain difference distribution. The blue line illustrated in Fig. 6(a) is the possible positioning distribution at the received signal power difference ΔP_{21} of -0.01, which is similar with that in Fig. 3(a). According to the signal power difference ΔP_{23} and ΔP_{31}, the possible positioning distributions can be generated as the green and red lines, respectively, as shown in Fig. 6(a). Therefore, the position of the optical receiver is the cross-point of these distribution lines. Actually, a cross area will be obtained owing to the deviation between the real environment and the ideal condition. Thus, the position of the optical receiver is estimated to be (402, 403) based on the center of the cross area for these distribution lines. The error distance is calculated to 3.6mm. We estimate the positions of 25 points on the optical breadboard to evaluate the performance of the indoor positioning estimation. Figure 6(b) shows the comparison of positioning distribution between the real position and the estimated position. The average distance error is less than 25mm and the maximum error is less than 68mm. The distance error at the corner of the optical breadboard is much higher than those of the center area, which may be caused by the longer transmission distance and the larger incidence angle. The maximum optical power is collected when the optical receiver is right below the LED transmitter. When the optical receiver locates far away the LED transmitter, the received signal power is smaller. The performance of the proposed visible light indoor positioning system can be further improved by optimizing the positioning algorithm, considering the non-LOS portion of the received signal power and precisely integrating the tilted stage and the rotation platform.

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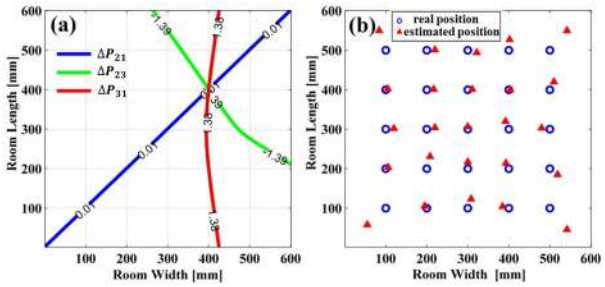


Fig. 6. (a) Positioning distribution lines when the real position is (400, 400), (b) the comparison of positioning distribution between the real position and the estimated position.

4. CONCLUSIONS

We propose a visible light indoor positioning system fashioned with a single LED transmitter and a single tilted optical receiver. The system offers a feasible way to make multiple measurements with different azimuth angles by rotating the tilted optical receiver. The indoor positioning algorithm is thus provided by angle gain difference. When the polar angle θ is 10°, the average distance error in the estimated position of 25 points is less than 25 mm in an indoor environment 600mm×600mm×1135mm. Experimental results are in good agreement with the simulations, suggesting that the proposed system is promising for visible light indoor positioning.

ACKNOWLEDGMENTS

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